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MILITARY SPECIFICATION

CLOTH, PLAIN WEAVE, POLYESTER, LOW AIR PERMEABILITY

This limited coordinated Military specification has been prepared by the U.S. Army Natick Laboratories (GL) based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL–C–43347A. It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

1. SCOPE

* **1.1** This specification covers polyester multifilament cloth.

* **1.2 Classification.** The cloth shall be of the following types as specified (see 6.2).

Type I — 8.5 oz.
Type II — 10.5 oz.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

CCC–T–191 — Textile Test Methods.
PPP–P–51 — Packaging, Packing and Marking of Textile Fabrics (Woolens, Worsteds, Cottons, Silks and Synthetics).

STANDARDS

FEDERAL

Fed. Std. No. 4 — Glossary of Fabric Imperfections.

MILITARY

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Ma- Materials ASTM D–276–62T: Tentative Methods for Identification of Fibers in Textiles.

(Copy of ASTM Standard may be obtained upon application to American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

Rules and Regulations Under the Textile Fiber Products Identification Act.

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D.C. 20580.)

3. REQUIREMENTS

3.1 Standard sample. The cloth shall be equal to or better than the standard sample

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(see 6.3) with respect to all characteristics for which the standard sample is referenced.

3.2 **Preproduction sample approval.** When specified (see 6.2), before production is commenced, a ¼-yard sample, full width of the cloth prior to treatment, and a 3-yard sample, full width of the finished cloth, shall be submitted or made ready for the contracting officer or his authorized representative, for inspection as specified in 4.2. The approval of the preproduction samples authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction samples shall be manufactured by the supplier in the same facilities to be used for the manufacture of the production item.

3.3 **Material.**

* 3.3.1 **Yarns.**

3.3.1.1 *Type I.* The yarns shall be polyethylene glycol teraphthalate, bright, high tenacity, multifilament. The warp shall be 220 denier (±3 percent), 2 ply for the plain weave; and 220 denier (3± percent), singles for the alternative weave (oxford). The filling shall be of 220 denier (± 3 percent), 3 ply. The yarns shall be tested as specified in 4.3.1.

3.3.1.2 *Type II.* The yarns shall be polyethylene glycol teraphthalate, bright, high tenacity, multifilament. The warp shall be 440 denier (±3 percent) single and the filling shall be of 1100 denier (±3 percent) single. The yarns shall be tested as specified in 4.3.1.

3.4 **Nonfibrous material.** Prior to the application of the water repellent treatment, the cloth shall not exceed 2.0 percent starch and protein content including chloroform-soluble and water-soluble material when tested as specified in 4.5.

3.5 **Color.** The color shall be natural.

* 3.6 **Physical requirements.** The finished cloth shall conform to the requirements shown in table I when tested as specified in 4.5.

TABLE I. *Physical requirements*

Type	Weight oz/sq. yd.	Yarns per inch minimum		Breaking strength minimum (Pounds)		Tearing strength minimum (Pounds)		Air permeability cu ft/min/sq. ft.[1]
	Minimum	Warp	Filling	Warp	Filling	Warp	Filling	Maximum
I	8.5	64[2]	43	475	540	25		2.0
II	10.5	70	35	550	750	30	35	2.0

[1] At a pressure drop across the cloth of 6 inches of water.
[2] The minimum warp texture for the oxford weave shall be 128 (see 3.6.1.2).

* 3.6.1 *Weave.*

3.6.1.1 *Type I and type II.* The weave for type I and II shall be plain (see 4.5).

3.6.1.2 *Type I alternate weave.* The alternate weave for type I may be woven with two warp ends weaving as one using a single 220 denier yarn. The minimum ends per inch shall then be 128. All other requirements for type I are applicable (see 4.5).

3.6.2 *Width.* The finished cloth shall be a minimum of 41 inches in width inclusive of selvages.

3.7 **Finish.** The cloth shall be scoured, high temperature heat set and calendered, and water repellent treated with a silicone emulsion, to result in a finish equal in character to that of the standard sample and to meet the test requirements of this specification (see 4.5, 6.3.1 and 6.4).

3.7.1 *Water repellency.* The silicone treated finished cloth shall show spray ratings no less than 90, 90, 90 when tested as specified in 4.5.

3.8 **Length and put-up.** Unless otherwise specified (see 6.2), the cloth shall be fur-

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nished in continuous lengths, each not less than 50 yards. Each length shall be put up on full width rolls as specified in PPP–P–51.

3.9 Fiber identification. Each roll of cloth shall be labeled, ticketed or invoiced for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act. (see 2.2)

3.10 Workmanship. The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards, when calculated as specified in section 4, shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Certificate of compliance.* Where certificates of compliance are submitted the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Preproduction sample inspection. When required, the preproduction sample of finished cloth submitted in accordance with 3.2 shall be visually inspected for appearance, color, and finish. The sample shall be tested for physical and chemical characteristics in accordance with table II and 4.5 to the extent applicable. The sample of cloth before treatment shall be tested for nonfibrous material in accordance with the method specified in 4.5.

4.3 Inspection. Inspection shall be in accordance with MIL–STD–105 except when otherwise indicated.

4.3.1 *Component and material inspection.* In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents. In addition to the quality assurance provisions of the subsidiary specifications, testing shall be performed on components and materials listed in table II for the test characteristics shown.

TABLE II. *Components.*

Characteristics	Requirement paragraph	Test methods
Material identification ...	3.3.1	[1]ASTM 276-62T
Denier	3.3.1 and 3.6.1.2	[1]4021
Luster	3.3.1	[1]
Tenacity	3.3.1	[1]
Ply		
Warp	3.3.1	[2]Visual
Filling	3.3.1	[2]Visual

[1] The supplier shall submit a certificate of compliance for these characteristics. When required, the certificate shall be accompanied by actual test, inspection, or other verifiable quality data.

[2] One determination shall be made from each sample unit and the results shall be reported as "pass" or "fail".

4.3.2 *Examination of the end item.* Examination of the end item shall be in accordance with 4.3.2.1 through 4.3.2.3.

4.3.2.1 *Yard-by-yard examination.* The required yardage of each roll shall be examined on the face side and all defects as defined in Section III of Fed. Std. No. 4 which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.3.2.1.1. No linear yard (increments of 1 yard on the measuring device of the inspection machine) shall be penalized more than 4 points. The sample size shall be 750 yards regardless of lot size. The sample yardage shall be examined from 25 rolls taken from 25 containers. An approximately equal number of yards shall be examined in each roll of the sample. The lot shall be unacceptable if the points per hundred square yards exceed 30.0 points. Point computation shall be as follows:

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$$\frac{\text{Total points scored in sample size} \times 3600}{\text{Contracted width of cloth (inches)} \times 750} = \text{Points per 100 square yards}$$

4.3.2.1.1 *Demerit points.* Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension ..... — one point
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension ..... — two points
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension ..... — three points
For defects exceeding 9 inches in any dimension ..... — four points

The following defects, when present, shall be scored four points for each yard in which they occur:

Baggy, ridgy, or wavy cloth
Objectionable odor
Width less than specified
Overall uncleanness or discoloration
Character of finish not equal to that of the standard sample

4.3.2.2 *Examination for length.*

4.3.2.2.1 *Individual rolls.* During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than 2 yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.3.2.2.2 *Total yardage in sample.* The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets. The rolls examined shall be those selected for the examination of individual rolls.

4.3.2.3 *Examination for compliance with the Rules and Regulations Under the Textile Fiber Products Identification Act.* During the yard-by-yard examination, each roll in the sample shall be examined for conformance to the Rules and Regulations Under the Textile Fiber Products Identification Act. Each roll not labeled in accordance with this act shall be a defect. The lot shall be unacceptable if two or more of these defects occur.

4.4 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP–P–51 to determine that packaging, packing and marking requirements of section 5 of this specification are complied with.

* **4.5 Testing of the end item.** The methods of testing specified in CCC–T–191, wherever applicable, and as listed in table III shall be followed. The physical and chemical values specified in section 3, except where otherwise specified, apply to the results of the determinations made on the sample unit for test purposes as specified in the applicable test method. All test reports shall contain the individual values utilized in expressing the final result. The sample unit shall be ¼ yard, full width of the cloth prior to the water repellent treatment for determination of the nonfibrous content, and 3 continuous yards full width of the finished cloth for the remaining tests. The sample size (number of sample units) shall be shown in the list below. The lot size shall be expressed in units of 1 yard. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified.

Lot size (yards)	*Sample size (units)*
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE III. *Methods of testing.*

Characteristics	Requirement paragraph	Test method
Unfinished cloth:		
Starch and protein content—including chloroform-soluble and water-soluble material..	3.4	2611
Finished cloth:		
Weight	3.6	5041
Yarns per inch		
Warp	3.6	5050
Filling	3.6	5050

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TABLE III. *Methods of testing (cont'd)*

Characteristics	Requirement paragraph	Test method
Weave		
Type I and type II	3.6.1.1	Visual[1]
Type I alternate	3.6.1.2	Visual[1]
Breaking strength		
Warp	3.6	5100
Filling	3.6	5100
Tearing strength:		
Warp	3.6	5132
Filling	3.6	5132
Air permeability	3.6	5450[2]
Finish	3.7	[3]
Water repellency	3.7.1	5526

[1] One determination shall be made from each sample unit and the result reported as "pass" or "fail".

[2] Test method 5450 shall be used except that a High Pressure Differential Air Permeability Machine shall be used. This machine shall be equipped with a manometer capable of measuring a pressure drop across the cloth under test of 1 to 21 inches of water and this manometer shall be graduated to read pressure in inches of water. In addition, the pressure drop across the orifice indicated by the vertical manometer shall be not less than 3 inches (A pressure drop of less than 3 inches across the orifice as indicated by the vertical manometer shall be permitted provided that at the time of testing the smallest possible orifice is being used) nor more than 26 inches. The determination shall be made at a pressure drop across the cloth of 6.0 inches of water.

[3] The supplier shall submit a certificate of compliance stating that the finished cloth has been high temperature heat set and was treated with a silicone emulsion.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2 and 6.2.1).

5.1.1 *Levels A and C.* The cloth, put as specified, shall be packaged in accordance with the applicable requirements for class 2 cloth of PPP–P–51.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2 and 6.2.1).

5.2.1 *Levels A, B, and C.* The cloth shall be packed in accordance with the applicable requirements of PPP–P–51.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP–P–51.

6. NOTES

* **6.1 Intended use.**

6.1.1 *Type I.* The type I cloth covered by this specification is intended for use as follows:

(a) The cloth for the slide fastener panels in Tent, Radome, Air Supported, Nike Hercules System, Single, Wall, 27 feet.

(b) The cloth for the lacing panel in Tent, Nike Hercules, Air Supported.

6.1.2 *Type II.* The type II cloth covered by this specification is intended for use in the construction of the Above-Ground Launchers and Radomes.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Type of cloth required (see 1.2).

(c) When preproduction sample is required (see 3.2).

(d) Minimum length if other than specified (see 3.8).

(e) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2.1 In the preparation of contracts or orders, it must be noted that various options, choices and alternatives, as indicated in PPP–P–51, may be exercised in the preparation for delivery of this cloth.

6.3 Standard sample. For access to standard sample, address the procuring office issuing the invitation for bids.

6.3.1 The standard sample was finished as follows (see 3.7):

High alkaline scour at a boil.
Dry.
Silicone emulsion applied.
Slack heat set at 410° F.
High temperature calendar 295° F. at pressure of 50 tons.

6.4 Effect of heat. It is to be noted that the polyester fibers available today vary in their shrinkage potential under the influence of heat. Based upon the supplier's choice of polyester yarn, appropriate time-temperature adjustments must be exercised (see 3.7).

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6.5 Marginal identification. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsover for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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